Updated: 06/2026

Yunpeng Zhang

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Education

University of Illinois Urbana-Champaign (UIUC) Aug. 2026 – Jun. 2028
M.Eng. in Electrical and Computer Engineering

China Jiliang University Sep. 2022 – Jun. 2026
B.Eng. (Honors) in Optoelectronic Information Science and Engineering **GPA: 93/100 | Rank: 1/223**

Internship

ByteDance | Data-aml | AI Automobile LLM/VLM Algorithm Intern Feb. 2026 – Jul. 2026

- Built an image captioning system for autonomous driving semantic understanding and cockpit HMI; vectorized images with an embedding model, selected cluster-center samples via **K-Means** for **SFT cold start**, and designed a two-stage **RLVR + GRPO** pipeline using JSON field-level TP/FN/FP matching and decoupled LLM-based VQA accuracy.
- Fine-tuned doubao-embedding-vision and seed-series VLMs for autonomous driving road element recognition; applied **FullSFT** and **LoRA** with positive/negative ratio ablations, **hard negative** construction, and model-forgetting detection; improved average Precision@K from 0.196 to **0.839**, achieving **59.73%** Micro-F1 and **54.67%** Macro-F1.
- Built an end-cloud offline LLM for always-on, wake-word-free intelligent cockpit; trained a high-precision, low-recall local rejection and **FunctionCall** parser with **SFT + ToolRL**, complemented by cloud-side high-recall rejection, two-stage LLM fallback, and Advisor arbitration; applied **strict/normal dual-mode** post-processing and **dynamic semantic library injection**, achieving **<100ms** latency, **95%** instruction accuracy, and **98%** rejection accuracy.

XPeng Motors | Intelligent Cockpit Center | VLM Foundation Model Intern Oct. 2025 – Jan. 2026

- Optimized an image captioning model for higher information density and fewer hallucinations; designed a **GRPO** training framework with **decoupled VQA rewards** and **RLVR** to maximize downstream VQA accuracy; built a dual-verification data pipeline to filter information leakage, enforce vision-dependent answering, and improve training data **signal-to-noise**.
- Improved VLM spatial relationship understanding and reasoning; built spatial benchmarks and aligned open-source model metrics, then applied **SFT cold start** and **GRPO** with custom spatial-reasoning rewards on curated data; achieved **SOTA** on CountBench and RefSpatial Bench.

Trusight Technology | Trusight AI Research | Computer Vision R&D Intern Jul. 2025 – Sep. 2025

- Optimized object tracking robustness under occlusion, fade-in/out, and tracking-loss scenarios; led the transition from Tracking-by-Detection → Tracking-by-Template → **Point tracking**; benchmarked, optimized, deployed CoTrack, TAP-Next, Locotrack in real-world scenarios; improved tracking accuracy by **10%+** and computational **efficiency** by **2x**.

Research Experience

3DMM-FLUX: Instruction-Guided Facial Expression Editing | Undergraduate Thesis
[Code] [Dataset] [🤗 Hugging Face] Mar. 2026 – May 2026

- Proposed an instruction-driven expression editing framework to address **identity drift** and **fine-grained control** in diffusion-based editing, using **FLUX** as the generative backbone and **DECA, FLAME, Nvdiffrast** for differentiable 3DMM control; designed a **Conditional DECA Encoder** and **RCG** inference strategy, achieving **AU Acc↑22.2pp to 81.4%**, **ID Sim↑0.072 to 0.824**, and **FID↓8.7 to 15.6**.

Research Assistant | Institute of Software, Chinese Academy of Sciences Sep. 2025 – Oct. 2025

- Analyzed AI Scientist, an automated scientific paper generation system, covering idea generation, novelty retrieval, automated experiments, paper writing, and review; clarified prompt structures and stage-wise interaction patterns.
- **AIGC Text Detection:** Built a **RoBERTa + AdaLoc** sentence-level detector with **sliding-window voting** and **overlapping-window aggregation**, achieved **92%** sentence-level accuracy in distinguishing human from AI.

Text Vectorization for Illegal Advertising Recognition Aug. 2024 – Sep. 2024

- Compared the recognition efficiency of **One-Hot**, **TF-IDF**, and **Word2Vec** text vectorization methods in long-text illegal advertising classification; contributed to 1 EI conference [Paper] at NLPIR 2024 and 1 [software copyright].

Competition Experience

Kaggle: Student Math Misconception Analysis | Silver Medal Sep. 2025

- Built a **MAP@3**-optimized misconception classifier for math problem responses; applied feature engineering, **classification-as-generation** reformulation, SFT, and weighted ensembling to fuse multi-model predictions; deployed GPTQ 4-bit quantized with vLLM for low-latency, high-throughput serving, achieving test-set score **↑20% to 97.83%**.

National Mathematical Modeling Competition | Provincial First Prize | Team Leader Sep. 2024

- Led the team to complete a crop planting strategy optimization problem, responsible for data preprocessing, building a multi-objective optimization model, and solving for the optimal planting plan.

China International College Students' Innovation Competition & National Undergraduate Physics Experiment Competition | Provincial Silver Awards | Team Leader May 2024 –May 2025

- Participated in the R&D of a power-compensation battery isothermal calorimeter, responsible for system construction and data processing. Authorized 3 [**software copyrights**], 1 invention [**patent**], and 1 utility model [**patent**].

Honors & Awards

- **Undergraduate Honors:** Outstanding Graduate, Honors Bachelor's Degree recipient, Merit Student (3×)
- **Scholarships:** 2025 Zhejiang Provincial Government Scholarship, 2026 Guangzhou High-Speed Railway Metrology Scholarship, First-Class School Scholarship (4×)

Skills

- **Languages:** Chinese (Native), English (Proficient), IELTS 7.0 (L6.5, R9.0, W7.0, S6.0), GRE 327 (V157, Q170, W3.5).
- **Programming:** Python, Matlab, C/C++
- **OS:** Linux, MacOS, Windows
- **Frameworks & Libraries:** Pytorch, Tensorflow, JAX, OpenCV, Swift, vLLM, DeepSpeed, Numpy, Pandas, Scikit-learn
- **ML/DL:** LLMs/VLMs Post-training, RL, RLVR, AIGC, Agentic RL, Quantization, Object Det/Trk/Seg.